

Thermodynamic State Vector Inventory Discrepancy Evaluator: Enforcing Conservation Laws in Biogeochemical Monad Simulations

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Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

Sprint 059 Technical Report

Abstract

Macroscopic Earth system models and complex biophysical simulations require strict adherence to physical conservation laws to maintain long-term stability and scientific validity. In Sprint 059, we introduce the **Thermodynamic State Vector Inventory Discrepancy Evaluator** (`src/thermodynamics/state_validator.ts`), a robust verification subsystem for the Web of Life framework. This module bridges macroscopic thermodynamic balance equations with discrete pool inventory accounting by evaluating absolute differences between actual stock transitions and expected flux-derived deltas. By programmatically enforcing the First Law of Thermodynamics (matter and energy conservation) and monitoring Second Law compliance (entropy and dissipation tracking), the validator acts as an immutable purity check within the simulation pipeline. We outline the mathematical formulation of stock deltas, the algorithmic architecture of the `StateValidator` class, and its integration within Earth pod telemetry to prevent unquantified thermodynamic drift.

1 Introduction & Systems Ecology Context

In ecosystem energetics and biogeochemical cycling, maintaining rigorous mass-energy balance is essential. Systems ecology models frequently suffer from numerical drift or unmodeled sinks and sources when tracking multi-pool interactions (Carbon, Nitrogen, Phosphorus, Water, and Energy stocks).

The Web of Life simulation architecture models these dynamics through discrete monad thermodynamic processes (`src/thermodynamics/monad_process.ts`) and state vectors (`src/thermodynamics/state_vector.ts`). To ensure that simulated ecosystems do not violate fundamental physical boundaries—specifically solar-input-only boundary conditions and matter conservation—Sprint 059 delivers an automated inventory discrepancy evaluator.

2 Mathematical Formulation

The evaluator operates on temporal transitions $t \rightarrow t + \Delta t$, comparing actual state variations against integrated flux accumulators for each pool i :

1. **Actual State Delta** ($\Delta S_{\text{actual},i}$):

$$\Delta S_{\text{actual},i} = S_i(t + \Delta t) - S_i(t) \quad (1)$$

2. **Expected Flux-Derived Delta** ($\Delta S_{\text{expected},i}$):

$$\Delta S_{\text{expected},i} = \sum_{j \in \text{Inputs}_i} \Phi_{j \rightarrow i} - \sum_{k \in \text{Outputs}_i} \Phi_{i \rightarrow k} \quad (2)$$

3. Absolute Discrepancy (ϵ_i):

$$\epsilon_i = |\Delta S_{\text{actual},i} - \Delta S_{\text{expected},i}| \quad (3)$$

4. Tolerance Validation Condition:

$$\text{violated}_i = \epsilon_i > \tau \quad (4)$$

where τ represents the system tolerance threshold (default: $\tau = 10^{-6}$).

3 Algorithmic Architecture & Implementation

Implemented in TypeScript within `src/thermodynamics/state_validator.ts`, the `StateValidator` class adheres to the `IStateValidator` interface. It aggregates pool keys, computes deltas, and constructs an immutable `DiscrepancyReport`:

```
export interface DiscrepancyReport {  
  timestamp: number;  
  totalDiscrepancy: number;  
  poolDiscrepancies: Record<string, {  
    actualDelta: number;  
    expectedDelta: number;  
    absoluteDifference: number;  
    violated: boolean;  
  }>;  
  withinTolerance: boolean;  
}
```

4 Thermodynamic Boundary Conditions

- **First Law Enforcement:** Total system discrepancy $\sum_i \epsilon_i$ must remain within tolerance τ . Non-zero drift signals unauthorized matter/energy leaks across the closed Earth system boundary.
- **Second Law Telemetry:** Violations trigger critical diagnostic events in the Earth Pod telemetry layer, forcing localization or halting divergent simulation trajectories.

5 Conclusion

Sprint 059 establishes rigorous thermodynamic oversight within the Web of Life platform. By continuously evaluating inventory discrepancies against integrated flux derivations, the system preserves mathematical consistency and physical realism across all biogeochemical monad processes.

Source code, test suites, and ongoing developments are available at the official repository: <https://github.com/pascalranoroariona/WebOfLife>.